

Knowledge Engineering Seminars

Ngô Minh Nhựt

2025

Outline

- ❑ **Course info**
- ❑ Research topics in Knowledge Engineering
- ❑ Literature review and situate research topic

Course info

☐ Materials

- Lecture notes
- Papers

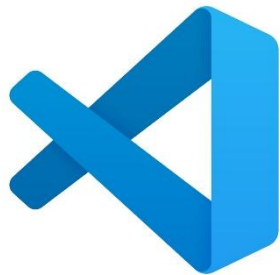
☐ Lecturers

- Dr. Ngô Minh Nhựt
- TA & Labs: Mr. Trần Hoàng Quân

Course info

❑ Projects and coding environment

- Python (with Visual Code)
- Google Colab (for projects)



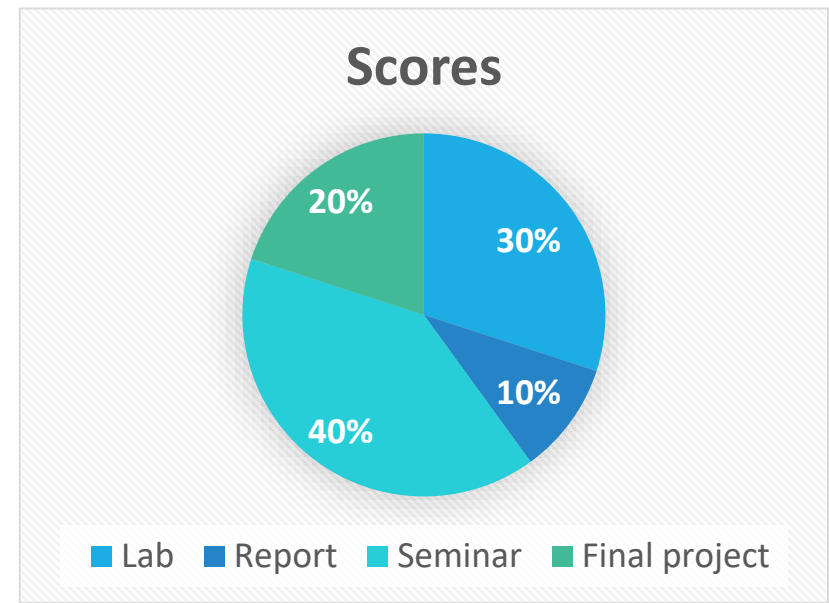
<https://code.visualstudio.com>



<https://colab.research.google.com>

Test and score

- ❑ Lab: 30%
- ❑ Report: 10%
- ❑ Seminar preparation: 10%
- ❑ **Final term**
 - **Seminar: 30%**
 - **Final project: 20%**



Score distribution is subject to change.

Test and score

❑ **Student's seminar: 30%**

- Mandatory, zero if missing
- *All students are required to participate in the seminars*

❑ **Final project: 20%**

- Mandatory, zero if failing to submit and participate Q&A

Outcome

- ❑ This course provides students with
 - Overview of research topics in knowledge engineering
 - Some skills necessary for doing research
 - Ability to solve problems
- ❑ This course provides students chances to
 - Study a research field and choose a problem of interest
 - Present the problem they study
 - Build applications in that field

Student seminar: students choose, prepare and present their topics

Course topic

No.	Topic
1	Information security
2	Machine learning and deep learning
3	Natural language processing
4	Speech technology
5	Literature review
6	Technical presentation

Course schedule

Week	Topic	Week	
1	Introduction	7	Student seminar #1
2	Topics for student seminars	8	Student seminar #2
3	Overview on Information Security	9	Student seminar #3
4	Overview on Machine learning	10	Student seminar #4
5	Technical Presentation	11	Final project
6	Introduction to Final project		

Course schedule is subject to change.

Seminar topic

No.	Topic	Description
NLP01	Natural language understanding	Help computers understand intent or meaning of a sentence
NLP02	Document classification	Classify text or document into categories
NLP03	Question Answering with RAG	Question answering system with Retrieval Augmented Generation
NLP04	LLM, GenAI and ChatGPT	Large language model, generative AI and ChatGPT
NLP05	Document summarization	Summarize text or documents into an abstract
SPE01	Large Audio Models	Apply concept of large language model from NLP to speech recognition
SPE02	Speaker recognition	Models for recognizing speaker from speech signals

Seminar topic

No.	Topic	Description
SEC01	Digital image forensics	Techniques for digital image forensic
SEC02	Security in sensor network	Attack vectors and solutions for them in sensor network
SEC03	Deep learning models for malware detection	Deep learning models for detecting malware
SEC04	Blockchain platforms and applications	Platforms for building decentralized applications
SEC05	Certificate Authority and how to build it	Certificate Authority and how to build it with OpenSSL and/or blockchain

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- ❑ Course info
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- ❑ Literature review and situate research topic

Research topics in KE

- ❑ Information security
- ❑ Machine learning and deep learning
- ❑ Natural language processing
- ❑ Speech technology
- ❑ Digital forensics

Information security

- ❑ A key field that helps secure information infrastructure of an organization, ensuring data communicated confidential and safe
- ❑ Confidential: prevent unauthorized access and use
- ❑ Safe: ensure data integrity, prevent unauthorized modification

Information security

□ Examples

- Encryption to ensure confidentiality
- Hashing and digital signature helps identify ownership of documents

Natural language processing

- ❑ A field that helps computers understand human languages and communicate with human
- ❑ Natural language differs formal language or programming language
 - Processing or understanding it is much more difficult

Natural language processing

- ❑ Processing, understanding then responding
- ❑ Examples
 - Intent classification: turn a question into an intent and variables
 - What time is it now in Tokyo?

Speech technology

- ❑ A study that helps understand spoken language and communicate with speech
- ❑ Available widely on digital devices nowadays

Speech technology

□ Problems

- Speech to text or speech recognition
- Text to speech or speech synthesis
- Speech transcription
- Speaker recognition

Digital forensics

- ❑ A branch of forensic science that focuses on recovery, investigation, examination, and analysis of data stored in digital devices often in relation to cybercrimes
- ❑ Most common application is to support criminal courts

Digital forensics

- ❑ Activities: capture, analysis, and preservation
- ❑ Areas of analysis: storage media, hardware and operating system, networks, and applications

Digital forensics

- ❑ Example: digital image forensics
 - Identify the provenance of an image, involving analyzing model of the camera that generates the image
 - Identify its integrity, examining if it has undergone one or more of adulteration processes

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- ❑ **Literature review and situate research topic**

Literature review

- ❑ An overview of the previously published works in a topic
 - Often includes a critical evaluation of the material. This is why we call literature review rather than literature report
- ❑ Usually aim to find gaps in the existing researches and find out research problems

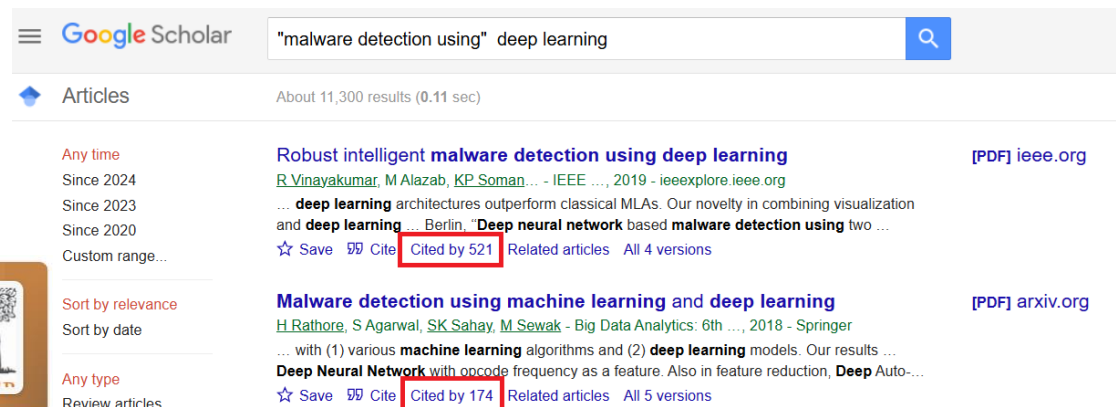
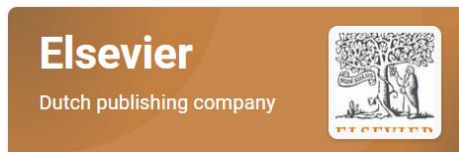
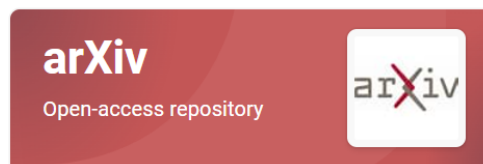
Literature review

- Key steps in literature review
 - Search for relevant literature
 - Google Scholar is a good starting source
 - Evaluate sources
 - Identify themes, debates, and gaps
 - Outline the structure
 - Write literature review

Literature review

□ Reputable sources

- Google Scholar
- Cornell University arXiv preprint
- IEEE Xplore
- Springer
- <https://paperswithcode.com>



Literature review

❑ Commonly unreliable sources

- Advertisement
- Social media
- Blog

Literature review

□ Tips for literature review

- Start with a bunch of seem-to-be related papers, e.g., 50 papers with title looking relevant
- Read briefly over the papers and select candidate ones, e.g., 10 papers
 - Sections to focus: abstract, introduction, and conclusion

Literature review

□ Tips for literature review (cont.)

- Read more carefully to find out most related ones, e.g., 5 papers
- Based on these papers, read carefully to study background, problems, approaches, solutions
- Analyze to figure out existing problems and gaps

Literature review

- ❑ Narrow down your topic
 - There are hundreds or even thousands of articles and books on most areas of study
 - It is important to go from broad area to a narrower one iteratively

Situate research topic

- Purpose of situating research topic is to determine
 - Where it lies in a given field
 - How it relates to other topics in the field
 - What research gaps it fills
 - What future research it indicates

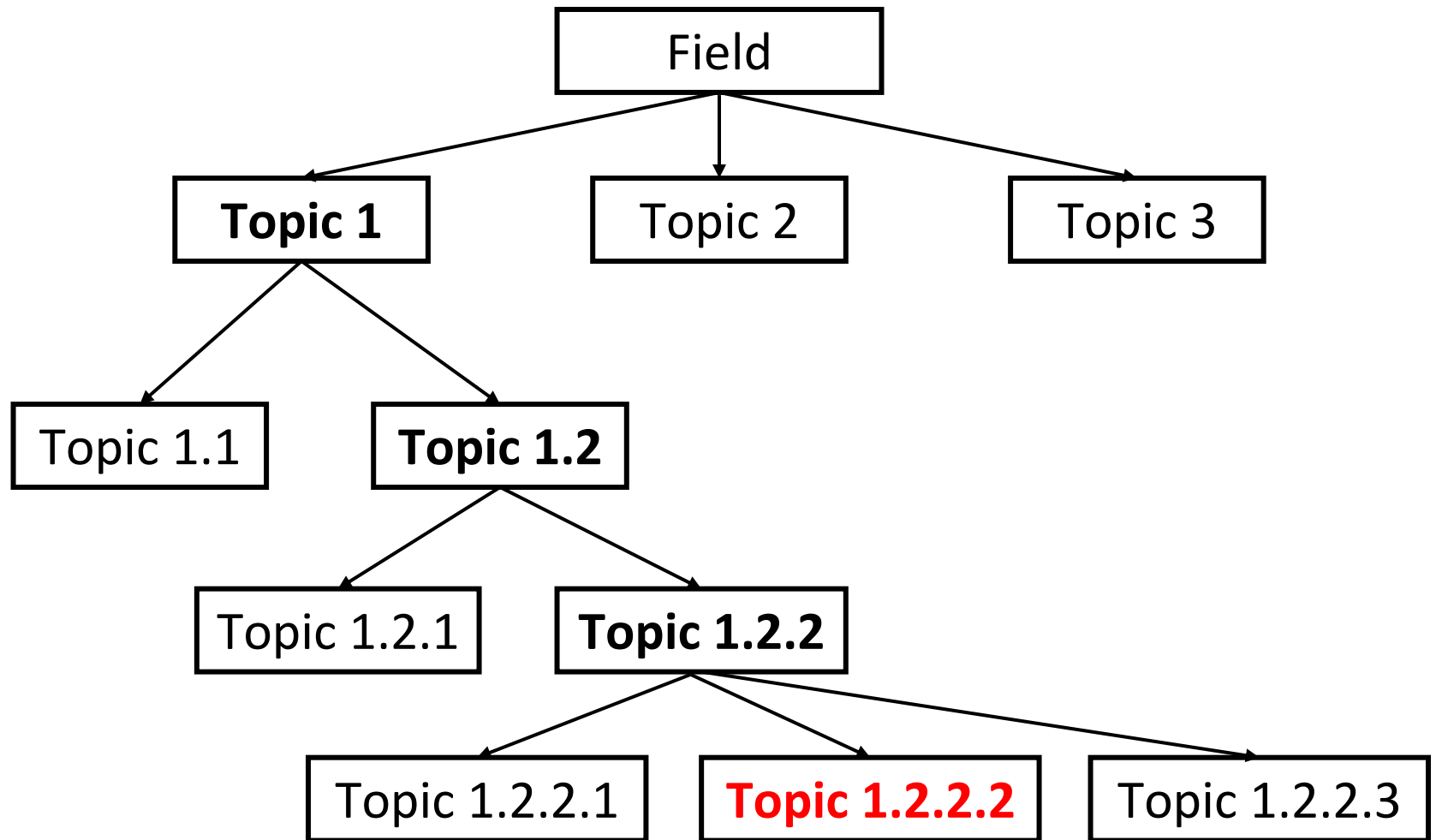
Situate research topic

- Strategy for situating research topic
 - Break down an interesting field into sub-topics
 - Identify preliminary research sub-topics based on your interest and experience in one or more sub-topics

Situate research topic

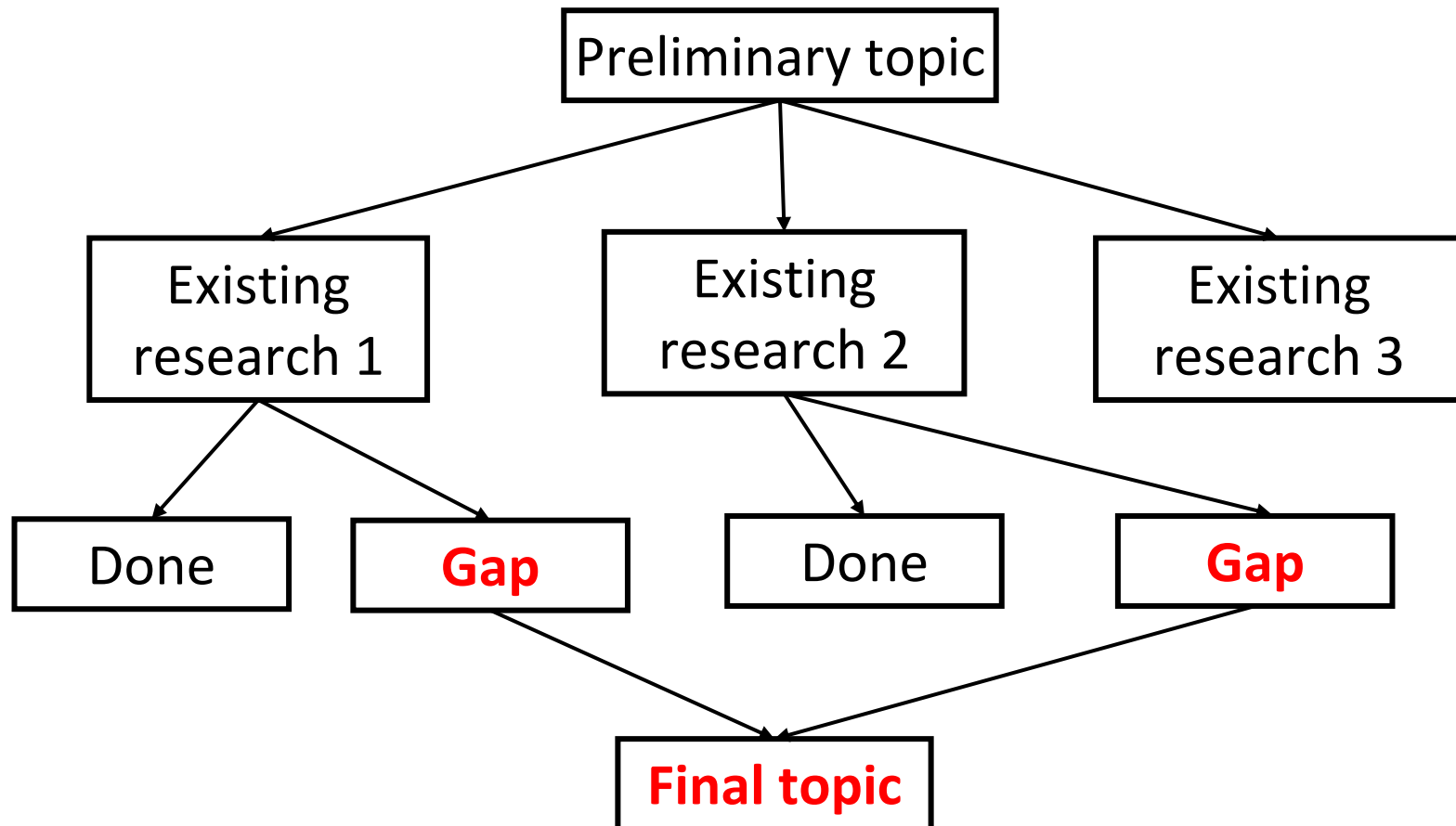
- Strategy for situating research topic (cont.)
 - Determine existing research, both accomplishments and gaps
 - Identify your final research topic based on your interest and experience in the sub-topics and on gaps in the existing research

Situate research topic



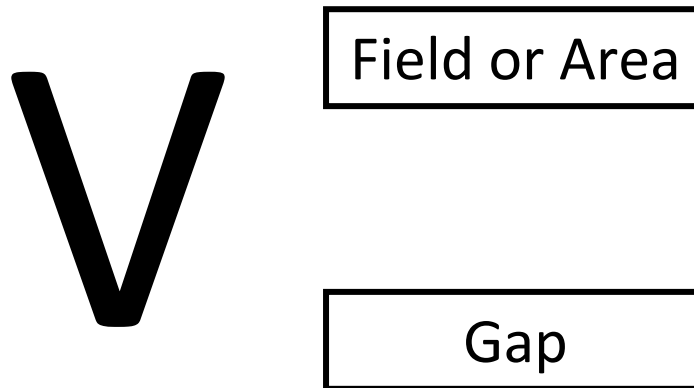
Situate research topic

- Choose your final topic



V-Structure for situating research topic

- ❑ Information about the final sub-topic and previous research on it
- ❑ Gap in the previous research
- ❑ Your chosen topic fills the gap in the previous research



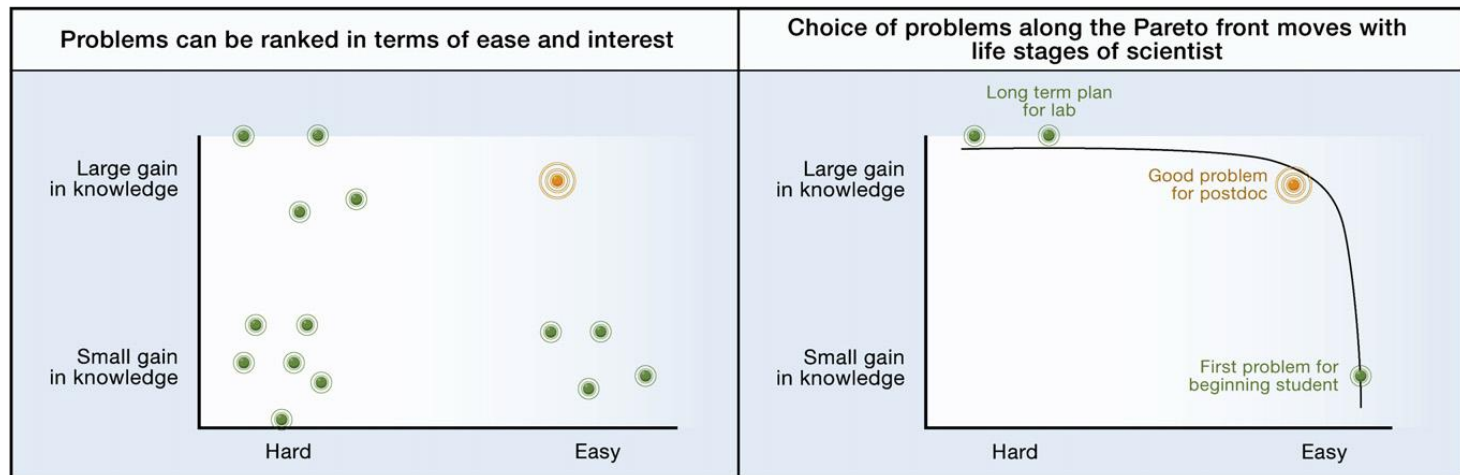
How to choose a good scientific problem

- Two dimensions of problem choice
 - **Feasibility:** How manageable the problem is, considering the lab's skills and technology.
 - **Interest:** The potential of the problem to generate new, valuable knowledge.

How to choose a good scientific problem

□ Two dimensions of problem choice

- Many problems are low-hanging fruits (easy but not very interesting). Some are difficult and of low interest.
- Ideally, one should aim for problems in the top-right
 - Both feasible and highly interesting.



Source: Uri Alon, *How To Choose a Good Scientific Problem*

Homework

- ❑ Read paper "*How to choose a good scientific problem*"
 - **There will be an online quiz**
- ❑ Formulate group
- ❑ Choose candidate seminar topics

References

- ❑ Prof. Jean-Christophe Terrillon (JAIST), *Research Communications course materials*
- ❑ Uri Alon, *How To Choose a Good Scientific Problem*